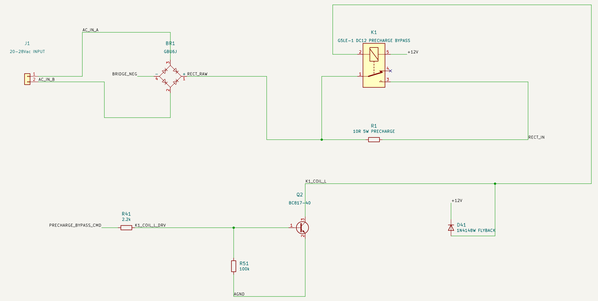
40 W CCM BOOST PFC | WIRED MODULAR LEARNING SCHEMATIC

Read the energy flow first (01 -> 02), then driver (03), sensing (04–06), fault logic (07), and MCU (08).

Same Rev8 circuit. Internal functional paths use wires. Matching port labels connect between modules; crossing wires have no implicit connection.

01 AC INPUT / RECTIFIER / PRECHARGE

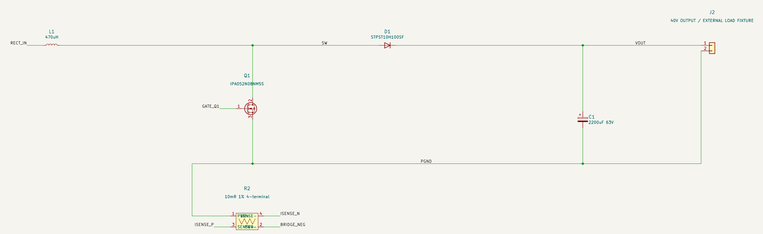
J1 -> BR1 -> R1 K1 bypasses R1 after precharge



K1 coil path is wired to Q2 and D41; its contact path bypasses R1.

02 BOOST POWER STAGE

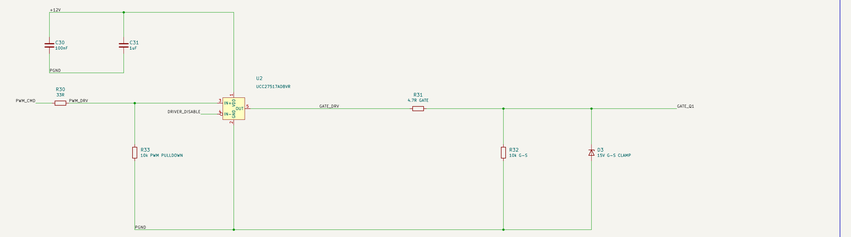
L1 -> SW -> D1 -> C1 / J2; Q1 and R2 complete the return



R2 pins 1/2: power return. Pins 3/4: independent Kelvin outputs to Module 04.

03 MOSFET GATE DRIVER

PWM_CMD -> R30 -> U2 -> R31 -> GATE_G1



IN– receives DRIVER_DISABLE. R33 is the existing hardware PWM pulldown.

04 KELVIN CURRENT SENSE

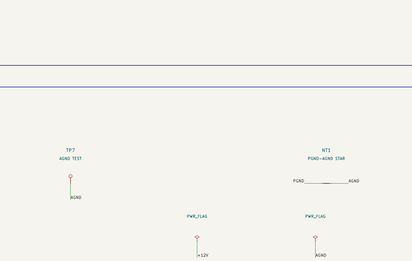
R2 sense terminals -> R20 / R21 -> U3 -> ISENSE_ADC



C20 is between INA_P and INA_N, not an input-to-ground capacitor.

05 VOLTAGE SENSE

Two separate resistor dividers: VOUT and RECT_RAW

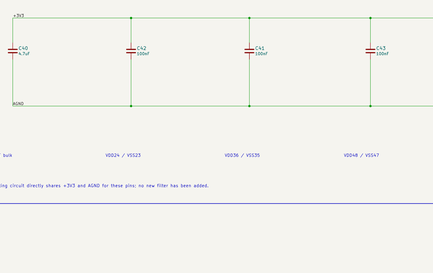


VOUT -> R10/R11/R12

RECT_RAW -> R13/R14/R15

06 HARDWARE OVERCURRENT

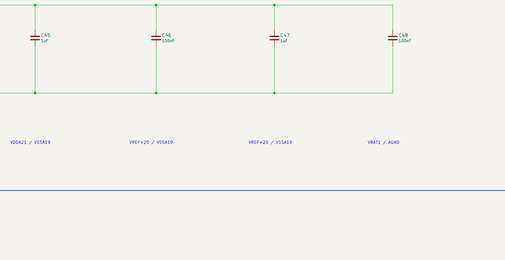
U4 compares ISENSE_ADC with OCP_REF



OC_RAW is an independent hardware shutdown input to Module 07.

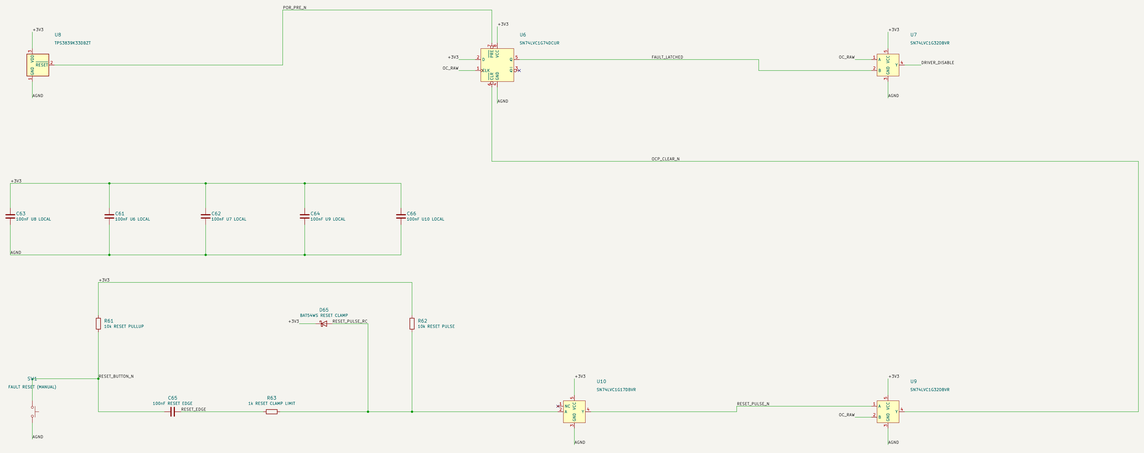
09 AUXILIARY POWER

J3 12 V auxiliary input -> U5 -> +3V3



07 FAULT LATCH / POR / MANUAL RESET

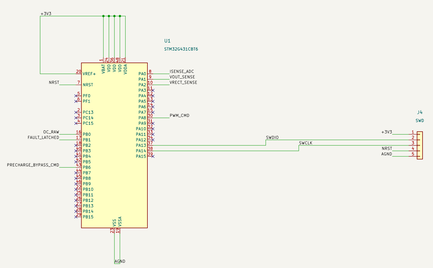
OC_RAW -> latch + fast OR -> DRIVER_DISABLE; no MCU clear



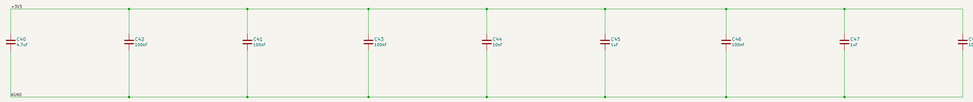
Manual reset edge -> RC/clamp -> Schmitt U10 -> OR U9 -> latch CLR.

08 STM32 CONTROL / SWD / DECOUPLING

Digital control interface: all existing supply and NC connections retained



MCU calculates digital PI / current reference / PWM. Fault status is read-only.

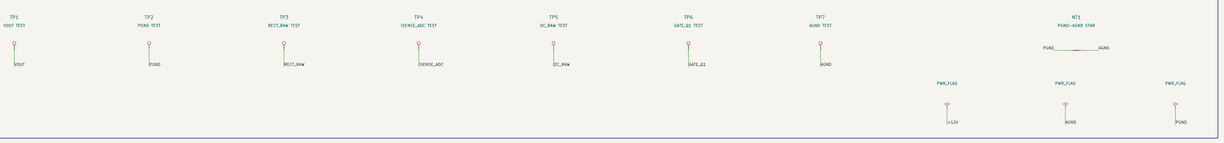


4.7 uF bulk | VDD2x / VSS23 | VDD36 / VSS35 | VDD48 / VSS47 | VDDA21 / VSSA19 | VDDA21 / VSSA19 | VREF+20 / VSSA19 | VREF+20 / VSSA19 | VBAT1 / AGND

The existing circuit directly shares +3V3 and AGND for these pins; no new filter has been added.

10 GROUND STAR / TEST POINTS

NT1 is the intended PGND / AGND connection; TP1–TP7 observe nets



RevB electrical topology unchanged

Sheet: /
File: BoostPFC_40W_WiredLearning.kicad_sch

Title: 40 W Boost PFC — Wired Learning Schematic

Size: A0 | Date: 2026-05-09 | Rev: Visual W1

KiCad 8.0.4 10.0.7 | Id: 1/1